Fireflies

Exercise 8.1: Fireflies (8P)

Imagine a huge cloud of fireflies (Glühwürmchen), each of them being at a fixed position $\vec{x}$ and randomly emitting light with propability $p(\vec{x})$.

You have made a snapshot with your camera, analyzing N of these fireflies in detail. Indexing them by $i = 1, \ldots, N$ you have produced a training data set $S = \{(\vec{x}_i, t_i)\}$, where $t_i = 1$ means that the firefly emits light and $t_i = 0$ means that it doesn't. Hence $p_i = \langle t_i \rangle$ is the probability of firefly i to emit light. Now you have trained a feed-forward network to predict this probability as a function of $\vec{x}$.

(a) Which activation function do you have to use in the last layer of the network? Explain your answer. (1P)

(b) Let $\{q_i\}$ the probability distribution predicted by the network. In machine learning, the *cross entropy* of the dataset S is defined as

$$H_S \;=\; -\frac{1}{N}\sum_{i=1}^{N}\Big(t_i \log_2(q_i) + (1-t_i)\log_2(1-q_i)\Big)\,,$$

where $\log_2$ denotes the logarithm to the base 2 and where the sum runs over all training examples. What would be the expectation value of $H = \langle H_S \rangle$ over infinitely many snapshots? (1P)

(c) Give a reasonable argument why for a sufficiently large training data set, it is reasonable to assume that $H_S \approx H$. (1P)

(d) Prove that H is minmal if the predicted probability distribution $\{q_i\}$ coincides with the true distribution $\{p_i\}$. (3P)

(e) Let H_{min} be the minimal value of H. Show that the difference $H - H_{min}$ gives just the so-called *Kullback-Leibler divergence* (see Wikipedia). (2P)

Exercise 8.2: Cross entropy (4P)

In the previous exercise, we considered probabilities of binary events (light or no light). Let us now consider M different possible events indexed by $\mu = 1 \ldots M$. Let us assume that these outcomes occur with the (true) probabilities $p_\mu(\vec{x})$. We would like to construct a network predicting these probabilities.

Please turn over ⇒

(a) Let $q_\mu(\vec{x})$ be the probabilities predicted by the network. Write down the cross entropy loss function. (1P)

(b) When minimizing the cross entropy, which constraint has to be taken into account? (1P)

(c) Show that the extremum again takes place at the point where the probability distributions p and q coincide. Use the method of Langrange multipliers to take care of the constraint in (b). (2P)

(Σ = 12P)

Please submit your solution electronically as a single PDF file until Wednesday, June 10, 2026, via WueCampus.